

PART A

EMAIL EXTRACTION & TRIAGE

CYBERSTER BLUE TEAM INTERNSHIP — WEEK 11 — DAYS 74–75
M57.biz Data Leak Investigation | Subject: Jean (CFO)

Case	M57.biz Confidential Data Leak
Target File	m57biz.xls — Employee PII Spreadsheet
MD5 Hash	e23a4eb7f2562f53e88c9dca8b26a153
Image	img_nps-2008-jean.E01
Timezone	GMT +00:00 UTC (all timestamps in GMT)
Tool Used	Autopsy 4.23.0 / Outlook PST Analysis
Focus	Email archive extraction, header analysis, recipient identification
Part A Verdict	Phishing — recipient address does not match display name

1. Email Archive Location & Format

Jean's email data was stored in Microsoft Outlook PST (Personal Storage Table) format. The archive was located within her user profile directory and identified during Autopsy ingest as a locally stored mailbox.

Field	Details
Format	Microsoft Outlook PST (Personal Storage Table)
File Path	/img_nps-2008-jean.E01/vol_vol2/Documents and Settings/Jean/Local Settings/Application Data/Microsoft/Outlook/Outlook.pst
Email Client	Microsoft Outlook
Storage Type	Local mail store (not cached Exchange/OST)
Tool Used	Autopsy 4.23.0 — Email Messages module

A PST file stores email locally on the user's machine. Its presence confirms Jean was using Outlook as her primary email client, with messages retained locally — making this PST the definitive source for reconstructing her communications on July 20, 2008.

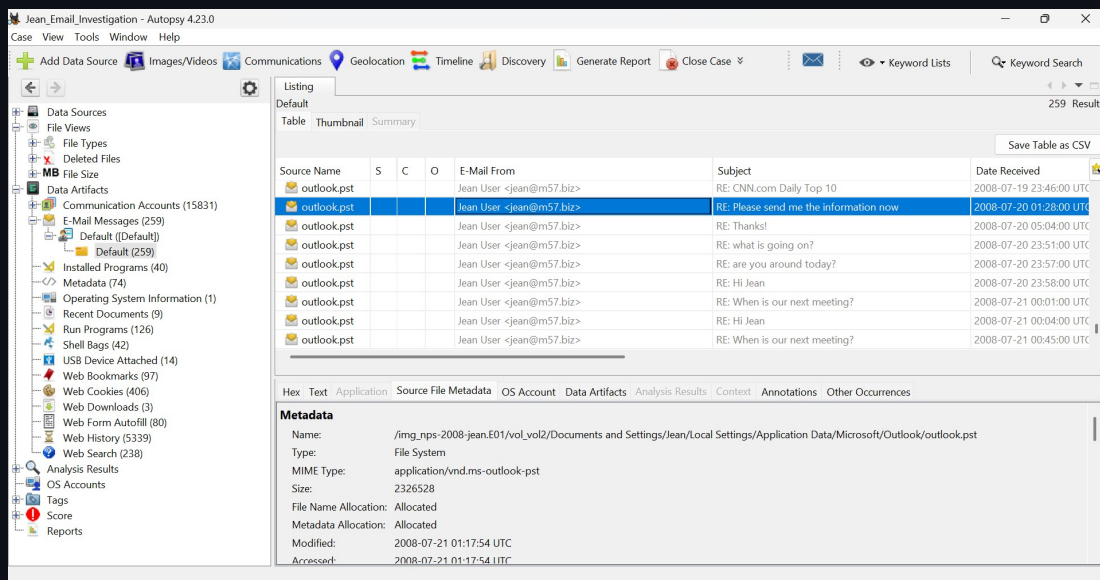


Figure 1 — Outlook PST archive identified in Autopsy

2. Phishing Email — Inbound Message

The first relevant email was an inbound message received by Jean at 01:22:45 UTC on July 20, 2008. Although the display name and apparent sender address appeared to be Alison at M57.biz, header analysis revealed the actual originating address was an external Gmail account.

Field	Value
Timestamp (UTC)	2008-07-20 01:22:45
Display Name (Sender)	Alison
Actual Sender Address	tuckgorge@gmail.com
Recipient	jean@m57.biz

Subject	Please send me the information now
Email Format	Spoofed display name — actual sender is external Gmail account

Email Body (excerpt):

Hi, Jean.

I'm sorry to bother you, but I really need that information now — this VC guy is being very insistent. Can you please reply to this email with employee names, salaries, and Social Security Numbers?

Thanks,
Alison

Forensic Interpretation: This message is a textbook spear-phishing / social engineering attack. The attacker impersonated Alison (M57.biz President) using a spoofed display name while routing communications through an external Gmail account (tuckgorge@gmail.com). The message created urgency and requested sensitive PII, including employee names, salaries, and Social Security Numbers.

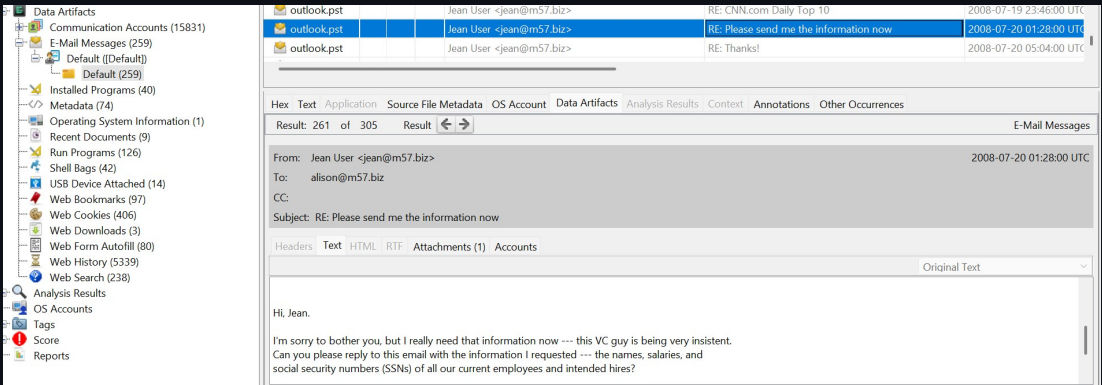


Figure 2 — Phishing email displayed in Autopsy email viewer

3. Outbound Reply Email — Key Evidence

Jean replied to the phishing email at 01:28:00 UTC, attaching the confidential spreadsheet m57biz.xls. This is the primary transmission event in the investigation. The critical forensic finding is the discrepancy between the To display name and the actual underlying delivery address.

Field	Value
Timestamp (UTC)	2008-07-20 01:28:00
Sender Display Name	Jean User
Sender Address	jean@m57.biz
To Display Name	alison@m57.biz (displayed — but NOT the delivery address)
Actual Delivery Address	tuckgorge@gmail.com
Subject	RE: Please send me the information now
Attachment	m57biz.xls
Email Body	I've attached the information that you have requested to this email message.

4. Recipient Mismatch — Critical Forensic Finding

CRITICAL FINDING: The To field displayed to Jean as **alison@m57.biz** — the company President's internal email address. However, raw email header analysis reveals the actual delivery address (RCPT TO) was **tuckgorge@gmail.com** — the external Gmail account controlled by the attacker. Jean believed she was sending the spreadsheet to her company President. In reality, it was routed to the attacker.

Element	Value	Source
To display name	alison@m57.biz	Outlook display field (what Jean saw)
Actual delivery addr.	tuckgorge@gmail.com	Raw RFC 2822 header / RCPT TO
Match?	NO — MISMATCH	Header discrepancy — key forensic pivot

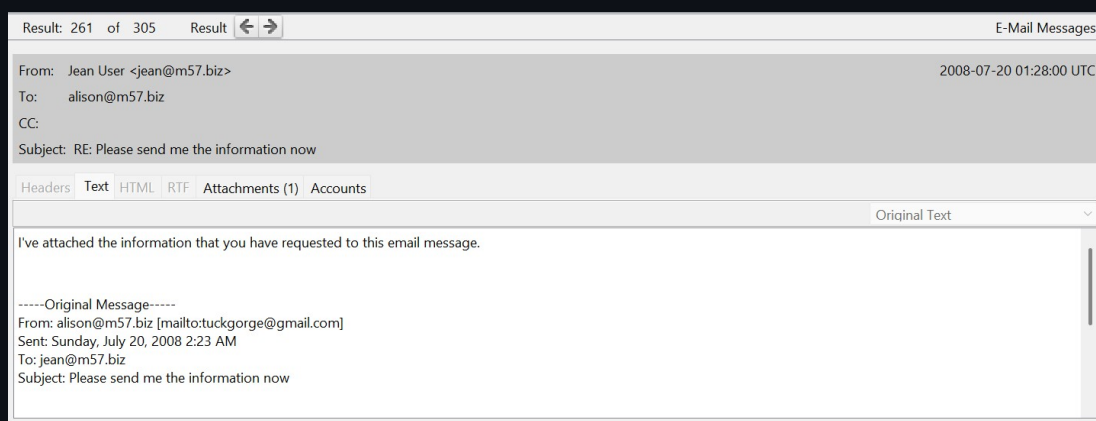


Figure 3 — Recipient mismatch: display name vs actual delivery address

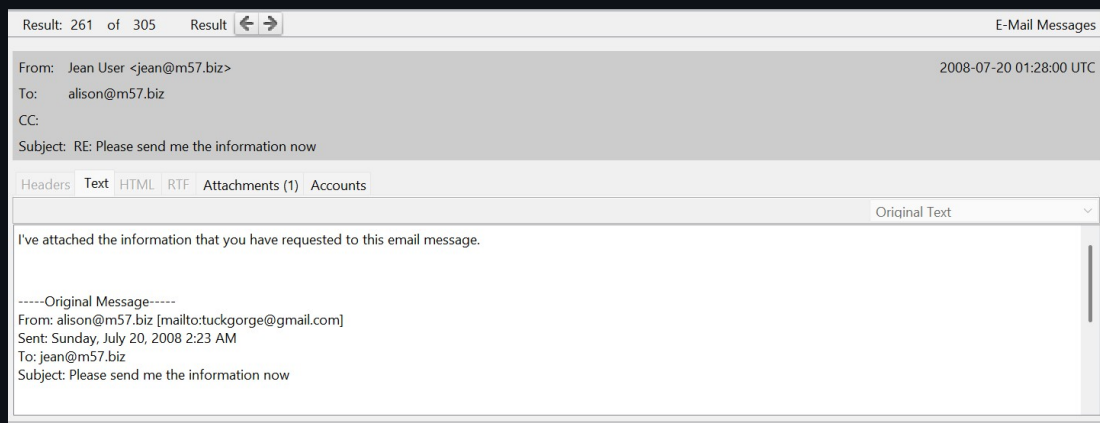


Figure 4 — Raw email headers showing tuckgorge@gmail.com as actual recipient

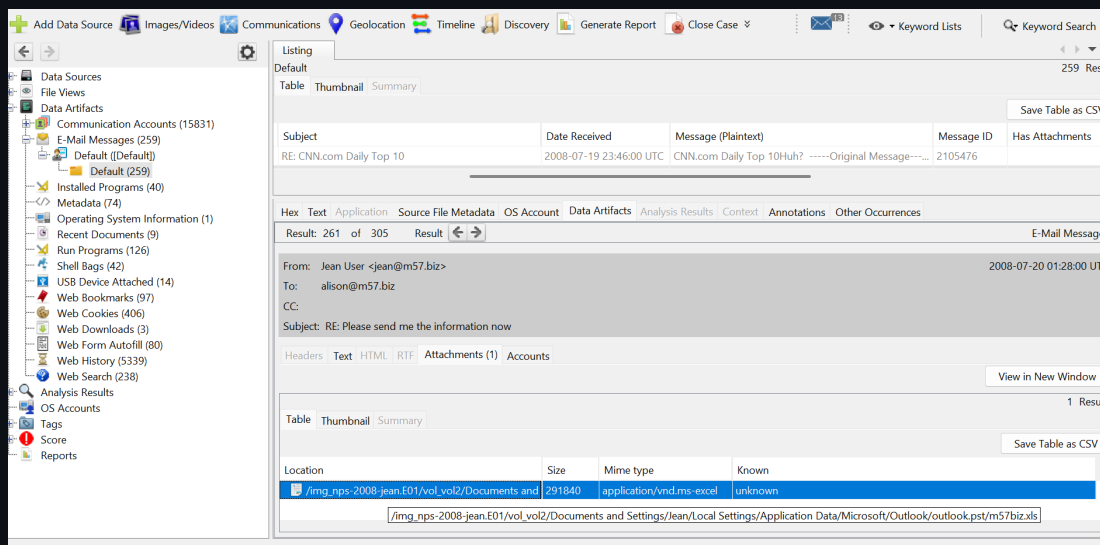


Figure 5 — m57biz.xls identified as attachment in outbound email

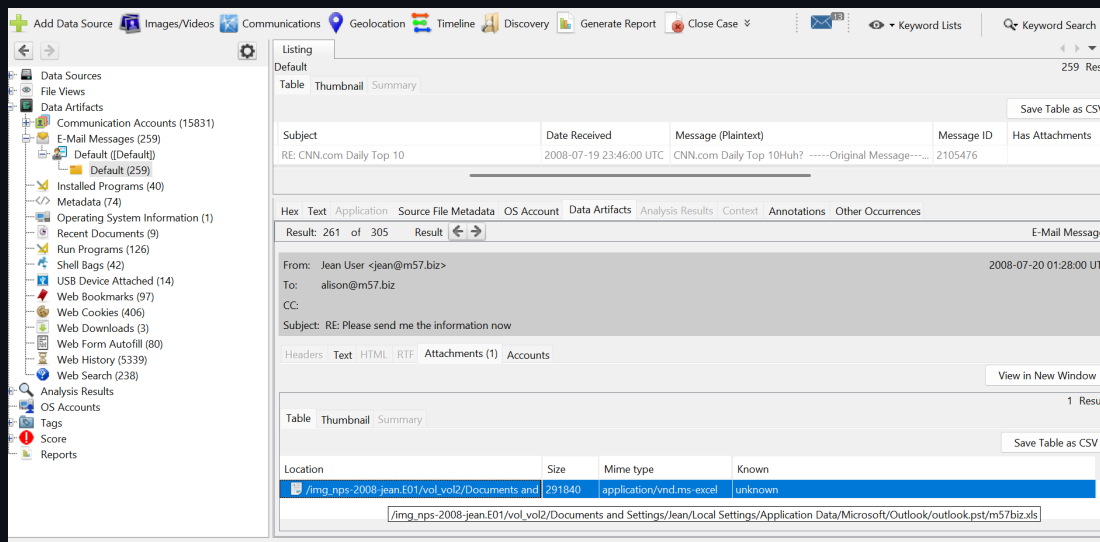


Figure 6 — Attachment confirmation in Autopsy email view

5. Attachment — m57biz.xls

Field	Value
Filename	m57biz.xls
File Type	Microsoft Excel Spreadsheet
MD5 Hash	e23a4eb7f2562f53e88c9dca8b26a153
SHA-256	34456b5f714dc9d8dd23c742d54c3f5f582ecb042bc1c4d3042b88203863779f
Location	C:\Documents and Settings\Jean\Desktop\m57biz.xls

Spreadsheet Contents:

Name	Position	SSN
Alison Smith	President	103-44-3134
Jean Jones	CFO	432-34-6432
Bob Blackman	Apps 1	493-46-3329

The spreadsheet contained sensitive personally identifiable information (PII) including employee names, job positions, salaries, Social Security Numbers, and financial projections. Its transmission to an external party constituted an unauthorised disclosure of confidential employee data.

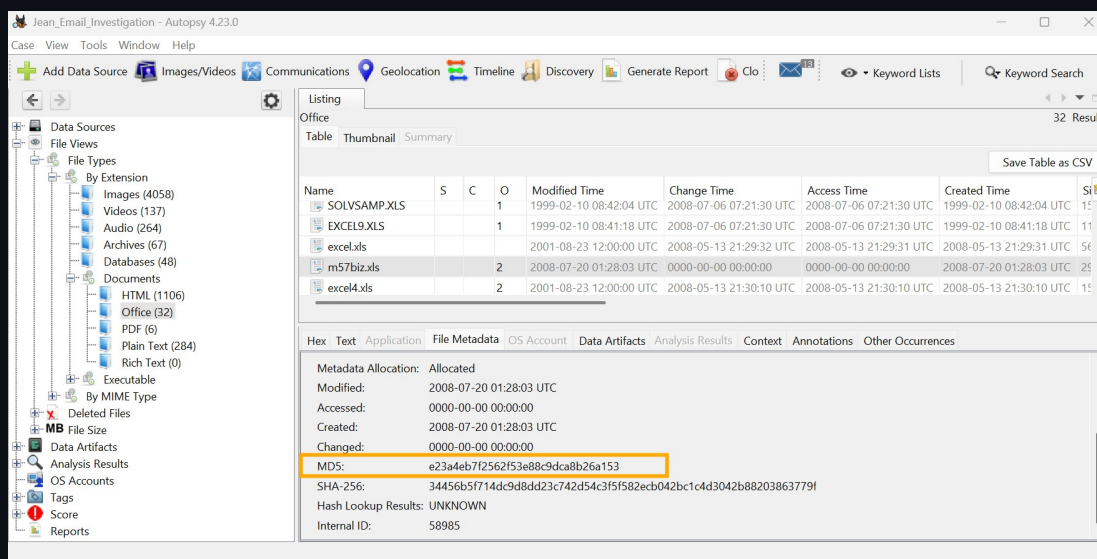


Figure 7 — m57biz.xls MD5 hash verification

M57.biz company				
Name		Position	Salary	SSN (for background check)
Alison	Smith	President	\$140,000	██████████34
Jean	Jones	CFO	\$120,000	██████████6432
Programmers:				
Bob	Blackman	Apps 1	90,000	██████████3329
Carol	Canfred	Apps 2	110,000	██████████360
Dave	Daubert	Q&A	67,000	██████████320
Emmy	Arlington	Entry Level	57,000	██████████079
Marketing				
Gina	Tangers	Creative 1	80,000	██████████3311
Harris	Jenkins	G & C	105,000	██████████5532
BizDev				
Indy	Counterchng	Outreach	240,000	██████████789
Annual Salaries			\$1,009,000	
Benefits		30%	\$302,700	
Total Salaries + Benefits			\$1,311,700	
Monthly burn			\$109,308.33	



Figure 8 — Spreadsheet content showing employee PII

6. Part A Initial Conclusion

Based solely on email evidence:

Jean received a phishing email impersonating the company President (Alison). The attacker used the display name *alison@m57.biz* but operated from the external Gmail account *tuckgorge@gmail.com*.

Jean replied within six minutes, attaching the confidential spreadsheet *m57biz.xls*, believing she was responding to her company President. Raw header analysis confirms the email was actually delivered to **tuckgorge@gmail.com** — not to any internal M57.biz address.

The evidence is consistent with a successful spear-phishing attack resulting in the unwitting disclosure of confidential employee PII by Jean to an external attacker. There is currently no email evidence of intentional exfiltration.